AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-13 | Entry window 13 October 2026 (full day) | 16 records screened in scope

16	9	2	1	4
RECORDS IN SCOPE (40 EXCLUDED)	OF 10 SUBTOPIC BANDS POPULATED	EFFECT ESTIMATES WITH A CI	TIER-A RECORDS	RECORDS CARRIED ON METADATA ALONE

Signal of the day

Prenatal ultrafine particles, not $PM_{2.5}$, track positive postpartum-depression screens in a 2,647-woman Shanghai cohort: OR 1.25 per IQR in the first trimester.

Residential trimester-specific UFP and $PM_{2.5}$; EPDS $\geq$10 at 6–10 weeks postpartum (20.4 % positive). First-trimester UFP **OR 1.252** (1.065–1.471) per IQR, whole-pregnancy **1.104** (1.047–1.164), both robust to $PM_{2.5}$ adjustment; $PM_{2.5}$ itself showed no stable independent association. It follows the 5 October UFP–dementia result (HR 1.22 per 10,000 cm^{-3}): two neuro endpoints in nine days carried by particle *number*, not mass. **Where it is weak:** an EPDS screen is not a diagnosis, and the abstract does not give the UFP surface's resolution or validation. UFP decays within a few hundred metres of roads; residential misclassification is probably large, and it is non-differential only if road proximity is unrelated to deprivation.

What changed today

- **Instrumentation is a third of the issue.** A polarisation-ceilometer aerosol typing algorithm (CONIOPOL, Vaisala CL61) validated over a year at Uccle; drone profiles showing when ground nodes represent the mixed layer; dust-lidar asymptotics implying classifiers under-count coarse absorbing dust; and a Taiwan PM_{10} forecaster in which *upwind observation density*, not trajectory sophistication, delivered the skill (RMSE 11.04 $\mu g\,m^{-3}$, 13 % below persistence).
- **Smoke source attribution in three regimes.** Fire-sourced $PM_{2.5}$ outperforms all-source mass for five cancer sites across China; fire-on/fire-off simulations put 17–21 $\mu g\,m^{-3}$ of 2024 São Paulo peaks on biomass burning, with every inventory short at the worst episode; PMF on 1,195 filters attributes 21–35 % of North Macedonian $PM_{2.5}$ to wood burning, >50 % on 83 % of Skopje episode days.
- **Particle counts are not microbes.** In dental ultrasonic scaling, sub-micron counts surged while culturable bacteria sat in the 1.1–3.3 μm stages — an optical counter's number signal is the wrong proxy for biological load.
- **One cross-day duplicate removed.** The Lancet Planetary Health multicountry admissions paper reached 13 October through the connector's `[EDAT]` bucket; it is carried once, in the 12 October issue.

Window continuity

The 12 October issue (built earlier in this run) closed with `last_entry_date` at 12 October. This issue collects **13 October in full**, continuous with it. 14 October follows as its own issue; 15 October is not opened under the 22:00 rule.

Corpus analytics

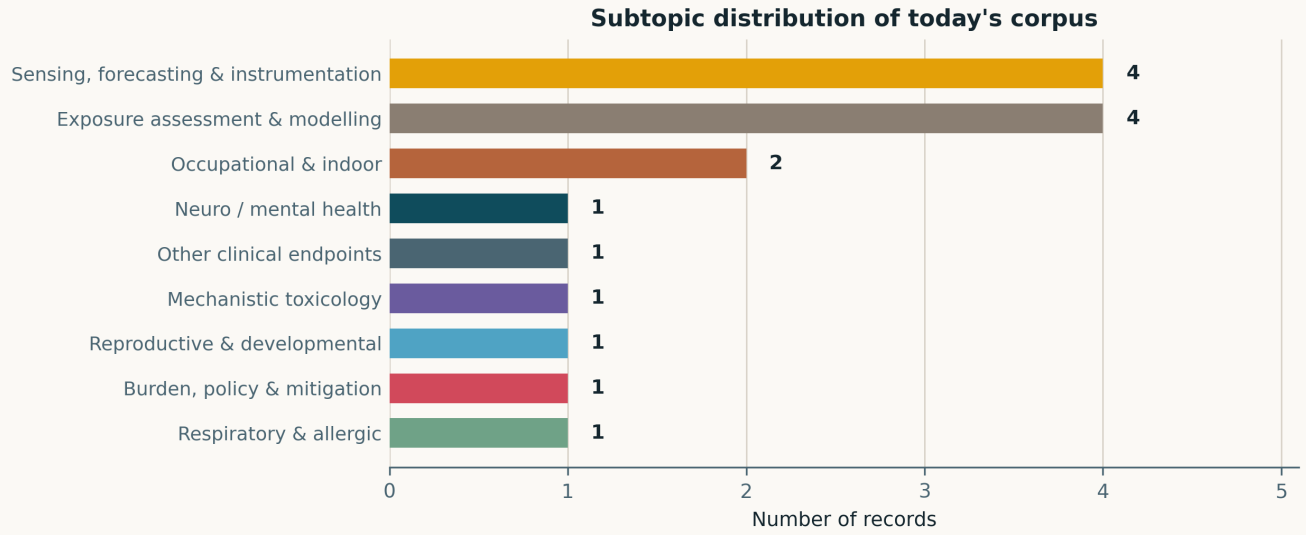


Fig. Subtopic assignment of the 16 in-scope records across nine bands. Sensing and exposure assessment tie at four each — the first issue since 8 October where measurement work equals the largest health band combined.

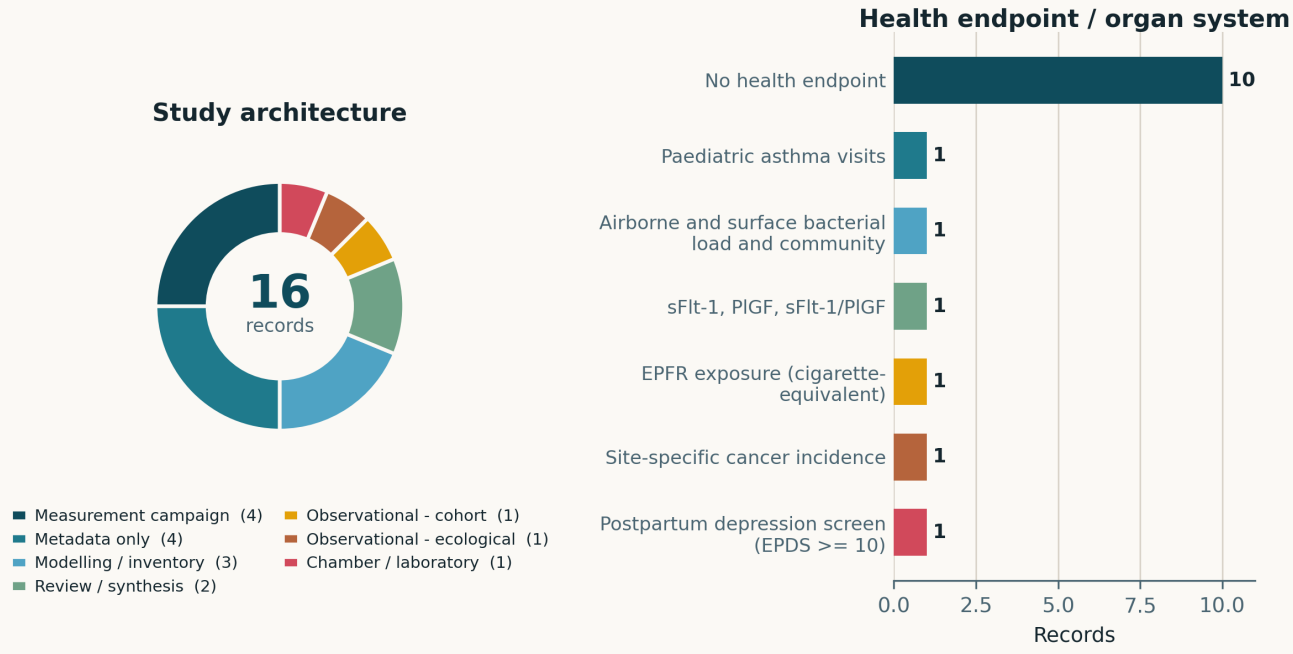


Fig. Left — architecture: four metadata-only records (25%), all Elsevier or Dove deposits without abstracts at harvest; one instrument development and one theoretical optics paper. Right — health endpoints are sparse: postpartum depression, cancer incidence, placental biomarkers, paediatric asthma.

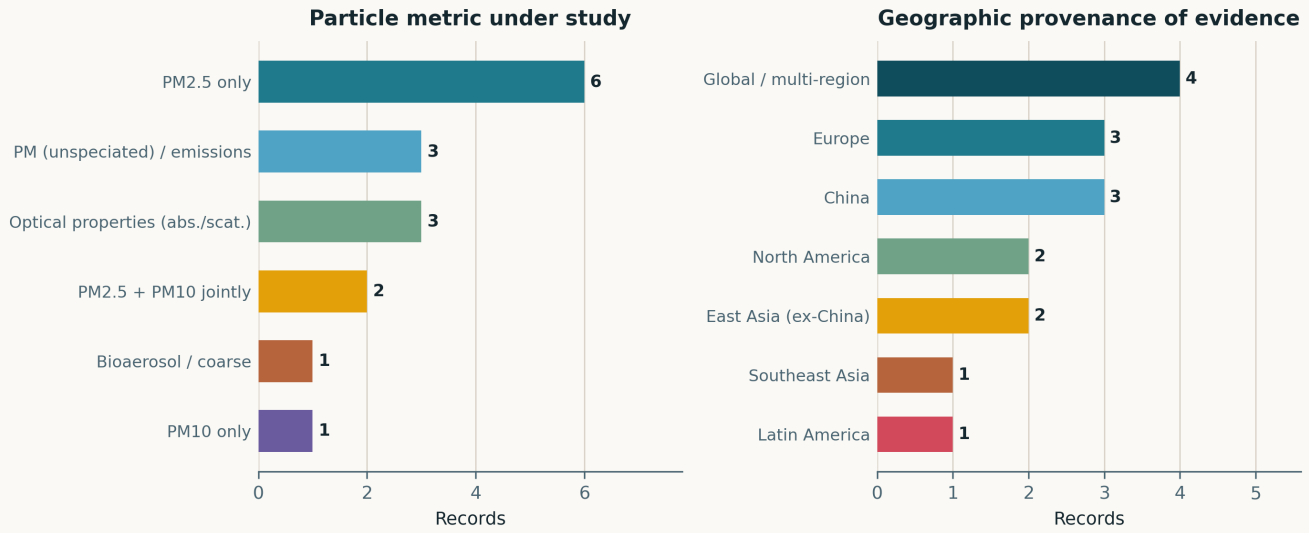


Fig. Left — particle metric: composition, source-resolved and vertical/optical quantities outnumber plain $PM_{2.5}$. Right — geography spans China, Taiwan, Vietnam, Brazil, North Macedonia, Belgium, Germany, Canada and the US.

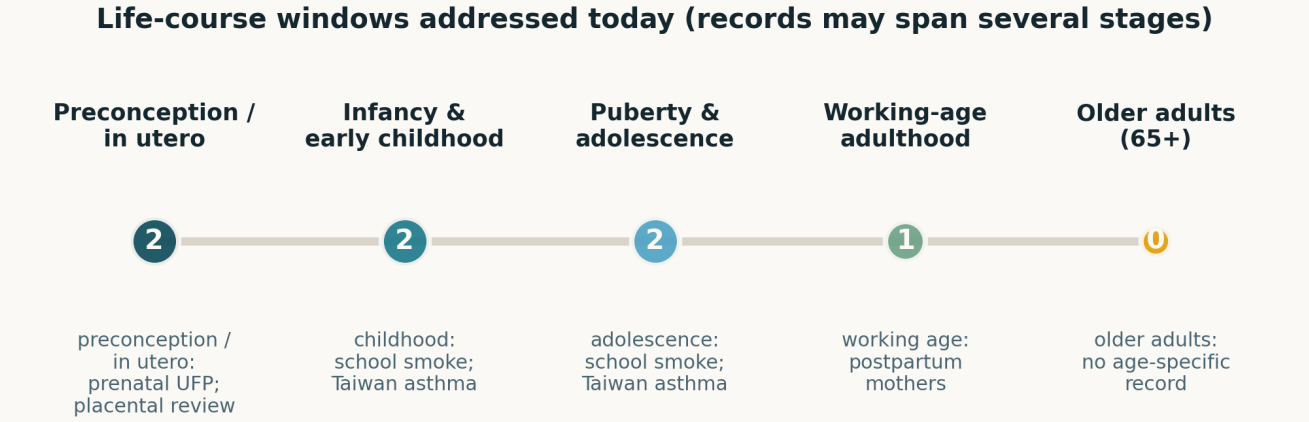


Fig. Life-course windows: in utero (prenatal UFP, placental biomarkers), childhood and adolescence (school smoke, Taiwanese asthma), working age (postpartum mothers). No older-adult record.

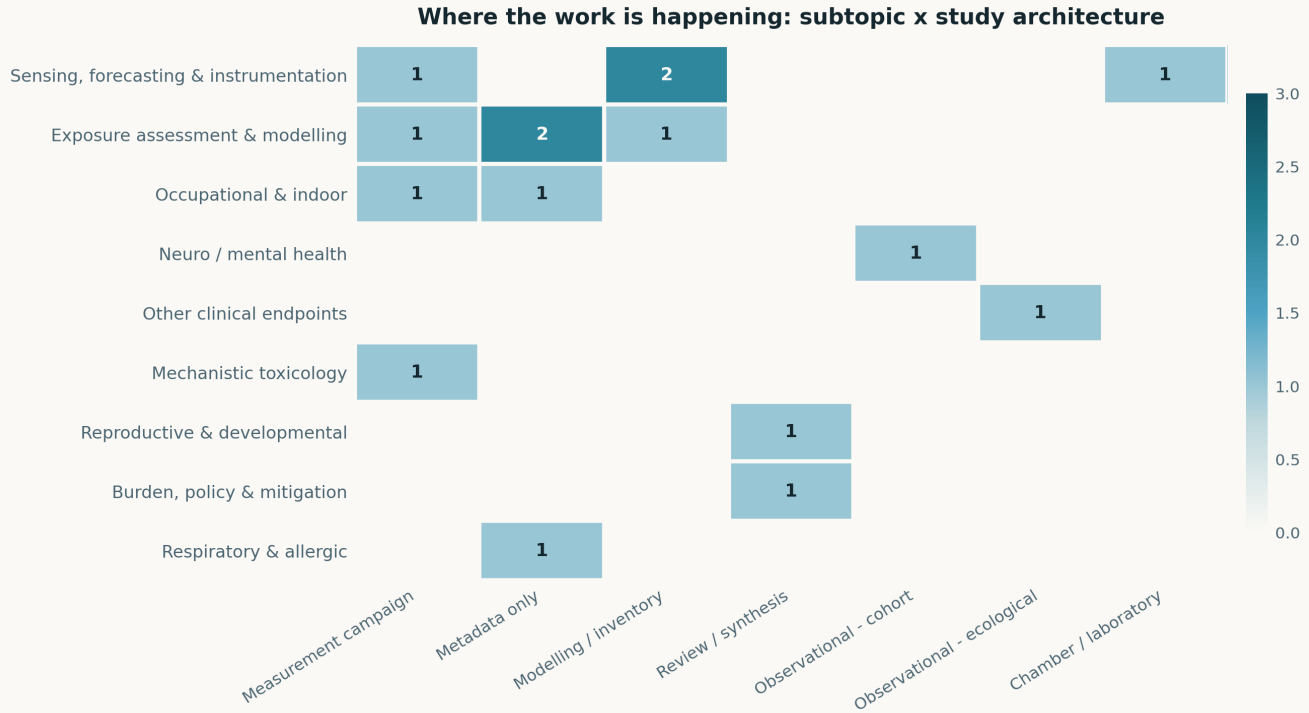


Fig. Subtopic × study architecture. The sensing row spans machine learning, instrument development, theory and field measurement — four architectures for four records.

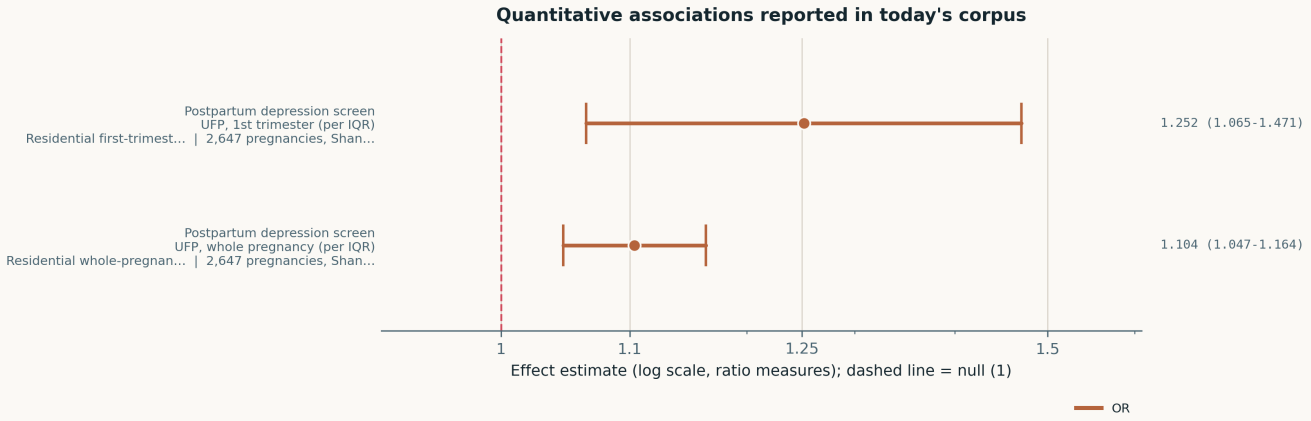


Fig. Two odds ratios per IQR of modelled residential UFP from the same 2,647 pregnancies; not independent. The first-trimester estimate is larger but wider; the whole-pregnancy estimate is the more stable of the two. No $PM_{2.5}$ estimate is shown because the abstract reports none as stable.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health — 1 record

Shanghai prenatal UFP postpartum depression 2026 Environ Pollut

Prenatal exposure to ultrafine particles and positive screening for postpartum depression: a prospective pregnancy cohort in Shanghai, China

2,647 pregnant women, two centres; 20.4% screened positive at 6-10 weeks postpartum. Per IQR, first-trimester UFP OR 1.252 (1.065-1.471) and whole-pregnancy UFP OR 1.104 (1.047-1.164); both survived adjustment for $PM_{2.5}$, which had no stable independent association. Stronger with male infants. Limitation: EPDS is a screen, not a diagnosis, and the UFP surface's spatial resolution and validation are not stated - UFP gradients decay over ~100-300 m from roads, so residential misclassification is likely large and non-differential only if traffic proximity is unrelated to deprivation, which in Shanghai it is not. Relevance: high - a second UFP-specific neuro signal this month that $PM_{2.5}$ mass does not carry, and that optical nodes cannot measure.

doi: 10.1016/j.envpol.2026.129330

Respiratory & allergic — 1 record

Taiwan greenness pollution child asthma 2026 J Asthma Allergy

Association of Urban Greenness and Air Pollution on Asthma Visits Among Children and Adolescents in Industrial and Non-Industrial Cities in Taiwan

METADATA ONLY. No abstract at harvest. Title indicates an analysis of greenness and air pollution with paediatric asthma visits contrasting industrial and non-industrial Taiwanese cities. Recorded for coverage. Relevance: low-moderate.

doi: 10.2147/jaa.s614875

Reproductive & developmental — 1 record

Placental angiogenic biomarkers review 2026 Reprod Toxicol

Exposure to environmental toxicants during pregnancy and placental angiogenic biomarkers: An integrative review

Whittemore-Knafl integrative review: 19 human studies after screening. Air pollutants predominantly showed an anti-angiogenic pattern - higher sFlt-1 or sFlt-1/PlGF and lower PlGF - as did PFAS, Cd and Cu; 14 of 19 studies reported at least one association in that direction. Limitation: 19 heterogeneous studies across four toxicant classes, no pooling or bias assessment, and 'at least one association' across multiple biomarkers and windows is a low bar. Relevance: low-moderate - plausible mechanism for PM-preeclampsia links, not evidence of it.

doi: 10.1016/j.reprotox.2026.109377

Mechanistic toxicology 1 record

EPFR size-resolved PM 2026 Environ Toxicol Chem

Size-Dependent Characteristics, Exposure Risk, and Formation Mechanisms of Environmentally Persistent Free Radicals in Atmospheric Particulate Matter

EPFRs were carbon-centred radicals adjacent to oxygen groups; $PM_{2.5}$ had the highest mass-normalised concentration (8.05e16 spins/g) and g-factor (2.00337). Cigarette-equivalent exposure was 0.326/day for $PM_{2.5}$ and 0.346/day for PM_{10}. Laboratory surrogates showed smaller particles generate more EPFRs owing to larger specific surface. Limitation: the cigarette-equivalent model is a communication device without dosimetric basis, and site/season details are absent from the abstract. Relevance: low-moderate - another surface-chemistry toxicity axis that mass sensing cannot see.

doi: 10.1093/etojnl/vgag274

Exposure assessment & modelling 4 records

North Macedonia biomass PMF 2026 Environ Pollut

Biomass burning drives urban $PM_{2.5}$ across the Western Balkans: evidence from a six-city source apportionment study in North Macedonia

1,195 filters at six urban-background sites over two year-long campaigns. Campaign means 29.1-62.2 $\mu g\,m^{-3}$ (1.2-2.5x the EU annual limit); residential wood combustion and open/waste burning resolved as distinct factors everywhere, together 21-35% of $PM_{2.5}$, with heating-season RWC up to 33x non-heating. Biomass exceeded 50% of $PM_{2.5}$ on 83% of Skopje episode days; an undocumented winter signal in Strumica fits greenhouse boilers. Limitation: PMF factor naming relies on tracer interpretation, and RWC/OBB separation without levoglucosan isotopes or 14C is assumption-laden. Relevance: high - a wood-smoke regime with bimodal evening peaks to 93 $\mu g\,m^{-3}$, where optical sensor response to fresh smoke is the calibration problem.

doi: 10.1016/j.envpol.2026.129339

Sao Paulo 2024 fire extremes 2026 J Environ Manage

Compound fire-meteorology extremes drove megacity air pollution during the 2024 South American fire season

Three multi-day 2024 episodes with daily $PM_{2.5}$ of 80-100 $\mu g\,m^{-3}$ in metropolitan Sao Paulo. Fire-off simulations missed the $PM_{2.5}$ and CO peaks; fire-attributable increments reached 17-21 $\mu g\,m^{-3}$ state-wide and a cumulative metropolitan burden of ~547 $\mu g\,m^{-3}$-day. Episodes differed in transport configuration, and the worst coincided with heat, dryness and weak ventilation. Limitation: all fire inventories underestimated the most extreme episode, so the attribution is a lower bound with unquantified error; a global model is coarse for a megacity receptor. Relevance: moderate - the inventory shortfall at the extreme is exactly where a dense surface network constrains attribution.

doi: 10.1016/j.jenvman.2026.131131

US school wildfire smoke trends 2026 Environ Res

Spatio-Temporal Trends in Wildfire Smoke Exposures at Schools and Childcare Centers in the United States and Implications for Child and Adolescent Health

METADATA ONLY. No abstract at harvest. Title indicates national spatiotemporal trends in smoke exposure at schools and childcare centres. Recorded for coverage; no estimate inferred. Relevance: high if the exposure surface is sensor-informed - school sensor deployments are the most common LCS siting in the US smoke literature.

doi: 10.1016/j.envres.2026.125883

Hanoi $PM_{2.5}$-bound PCDD/Fs 2026 Atmos Pollut Res

Levels and contributing sources to $PM_{2.5}$-bound PCDD/Fs and dl-PCBs in ambient air of Hanoi

METADATA ONLY. No abstract at harvest. Title indicates measurement and source attribution of particle-bound dioxins, furans and dl-PCBs in Hanoi $PM_{2.5}$. Recorded for coverage. Relevance: low - trace organics require filter sampling.

doi: 10.1016/j.apr.2026.103235

Sensing, forecasting & instrumentation 4 records

Taiwan transboundary PM_{10} forecast 2026 Environ Pollut

Forecasting transboundary PM_{10} pollution in northern Taiwan using upwind observations and transport timing

Upwind-transition-receptor framework with temporally separated training and evaluation for 1-4-day PM_{10} forecasts. Nested models attribute most added skill to distributed upwind observations; trajectory timing adds a smaller, receptor-dependent refinement. Best model (CatBoost with upwind data): RMSE 11.04 $\mu g\,m^{-3}$, 13.0% below persistence. Limitation: 13% over persistence at 1-4 days is modest, and upper-tail precision is limited by event scarcity - the episodes that matter are the ones the model sees least. Relevance: high - quantitative evidence that upwind observation density, not trajectory sophistication, buys forecast skill; a siting argument for upwind nodes.

doi: 10.1016/j.envpol.2026.129340

CONIOPOL CL61 aerosol typing 2026 Atmos Meas Tech

New operational perspective to identify aerosols and hydrometeors in real-time with a pioneering algorithm (CONIOPOL) based on single-wavelength polarization lidar (CL61)

Automated classification from a polarised automatic lidar-ceilometer combining depolarisation, attenuated backscatter and cloud base to separate aerosol, cloud phase and precipitation and to type aerosol as dust, smoke/marine, continental or mixed. One year at Uccle showed strong temporal and vertical correspondence with CAMS for transported dust and smoke. Limitation: single wavelength cannot separate smoke from marine aerosol, and validation against CAMS is model-conditioned agreement rather than in-situ truth. Relevance: high - ceilometers are already networked across Europe; vertical typing tells a surface sensor network when its column assumption is broken.

doi: 10.5194/amt-19-6415-2026

Dust lidar asymptotics 2026 Atmos Chem Phys

Asymptotic behavior of lidar scattering properties of absorbing dust aerosols across Rayleigh and geometrical-optics regimes: theory and implications

TAMUdust2020 and triaxial-ellipsoid calculations give shape-insensitive limits: Rayleigh S ~ x^-3, SSA ~ x^3, small depolarisation; geometric-optics depolarisation falls, SSA -> 1/2 and lidar ratio rises with strong absorption. Large, strongly absorbing dust can therefore mimic weakly depolarising classes and bias classifiers toward under-counting coarse and super-coarse dust. Limitation: theory only, no field test. Relevance: moderate - the same physics explains why optical PM sensors under-read coarse absorbing dust; the bias direction carries over.

doi: 10.5194/acp-26-14487-2026

Drone boundary-layer profiles Germany 2026 Atmos Chem Phys

Influence of irradiance and mixing layer height on the vertical trace matter distribution in the lower planetary boundary layer - drone-based investigation

Two 2.5-week summer campaigns: night profiles in the lowest 120 m showed multi-layer fine-scale stratification; O_3 and potential temperature were the most reliable mixing-height markers, and trace species outperformed several meteorological variables. Local sources and sinks affected in-layer distribution more than irradiance, but aerosol concentrations homogenised under strong radiation even at low mixing heights. Limitation: rural summer only; drone-borne particle sensors are themselves low-cost optical units with flow and RH artefacts that the abstract does not address. Relevance: high - directly tests when a ground node represents the mixed layer.

doi: 10.5194/acp-26-14417-2026

Occupational & indoor 2 records

Dental scaling bioaerosols 2026 J Hosp Infect

Divergent bacterial signatures in dental clinic bioaerosols: a real-field investigation

Ultrasonic scaling caused a large surge in sub-micron particle counts, but culturable bacteria concentrated in the 1.1-3.3 um stages. Airborne bacterial communities differed from PPE (ANOSIM R=0.633, p=0.001) and were indistinguishable from non-touch surfaces; oral pathogens were rare in air. Limitation: single clinic and procedure type, small sample count not given. Relevance: moderate - a clean demonstration that particle number from an optical counter is not a proxy for biological load; mirrors the 4 Oct suction study.

doi: 10.1016/j.jhin.2026.10.003

Global IAQ open standards framework 2026 Build Environ

A unified framework for global indoor air quality: transitioning from guidelines to actionable open regulatory standards

METADATA ONLY. No abstract at harvest. Title indicates a proposal to move indoor air quality from guideline values to open, enforceable standards. Recorded for coverage. Relevance: moderate - enforceable indoor standards would make low-cost indoor PM monitors compliance instruments, with the performance-certification burden that implies.

doi: 10.1016/j.buildenv.2026.115384

Burden, policy & mitigation 1 record

Canada wildfire smoke guidance 2026 Can J Public Health

Public health guidance about wildfire smoke: Four targets for improvement

Commentary proposing four distinct targets for improving Canadian wildfire-smoke guidance: a stronger evidence base (subpopulations, long-term effects, protective interventions), ethically deliberated recommendations, evaluation of guidance effectiveness beyond self-report, and a process-and-content standard for authoritative guidance. Limitation: no data. Relevance: moderate - the evaluation target presupposes measured indoor and personal exposure during smoke events, which is where low-cost sensors already dominate the evidence.

doi: 10.17269/s41997-026-01267-x

Other clinical endpoints 1 record

China fire-sourced $PM_{2.5}$ cancer 2026 Environ Pollut

Extreme, Accumulated, and Lagged Exposure to Fire-Sourced $PM_{2.5}$ and Site-specific Cancer Risk in China

Fire-sourced $PM_{2.5}$ was positively associated with several site-specific cancers, with lagged and accumulated metrics showing the broadest pattern (thyroid, kidney, breast, lung, leukaemia, lymphoma) and extreme exposure the narrowest (strongest for larynx). Against all-source $PM_{2.5}$, fire $PM_{2.5}$ associations were stronger for thyroid, kidney, larynx, lymphoma and oral cancers, and stronger in the north and in low-GDP areas. Limitation: no effect sizes, n or geographic unit in the abstract; many sites x three metrics is a multiple-comparison surface, and thyroid cancer incidence is notoriously driven by screening intensity, which tracks region and income. Relevance: moderate - source-resolved smoke again outperforms mass.

doi: 10.1016/j.envpol.2026.129343

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Shanghai prenatal UFP postpartum depression	Prenatal exposure to ultrafine particles and positive screening for postpartum depression: a prospective pregnancy cohort in Shanghai, China	*Environ Pollut*	Prospective cohort	Residential trimester-specific UFP number and $PM_{2.5}$ (modelled)	China (Shanghai)
Reproductive & developmental					
Placental angiogenic biomarkers review	Exposure to environmental toxicants during pregnancy and placental angiogenic biomarkers: An integrative review	*Reprod Toxicol*	Integrative review	$PM_{2.5}$, PM_{10}, NO2, O3 among other toxicants	Global / multi-region
Respiratory & allergic					
Taiwan greenness pollution child asthma	Association of Urban Greenness and Air Pollution on Asthma Visits Among Children and Adolescents in Industrial and Non-Industrial Cities in Taiwan	*J Asthma Allergy*	Metadata only (no abstract)	Ambient air pollution (unspecified) and greenness	Taiwan
Mechanistic toxicology					
EPFR size-resolved PM	Size-Dependent Characteristics, Exposure Risk, and Formation Mechanisms of Environmentally Persistent Free Radicals in Atmospheric Particulate Matter	*Environ Toxicol Chem*	Field measurement	EPFRs (EPR) in $PM_{2.5}$, PM_{10}, TSP; SiO2-catechol-ZnO simulations	China
Occupational & indoor					
Dental scaling bioaerosols	Divergent bacterial signatures in dental clinic bioaerosols: a real-field investigation	*J Hosp Infect*	Field measurement	Optical particle sizer; Andersen six-stage culturable and filter bioaerosol sampling	Not stated (dental clinic)
Global IAQ open standards framework	A unified framework for global indoor air quality: transitioning from guidelines to actionable open regulatory standards	*Build Environ*	Metadata only (no abstract)	Indoor air quality metrics (incl. PM)	Global / multi-region
Sensing, forecasting & instrumentation					
CONIOPOL CL61 aerosol typing	New operational perspective to identify aerosols and hydrometeors in real-time with a pioneering algorithm	*Atmos Meas Tech*	Instrument development	Volume depolarisation ratio and attenuated backscatter (Vaisala CL61 ceilometer)	Belgium (Uccle)
Drone boundary-layer profiles Germany	Influence of irradiance and mixing layer height on the vertical trace matter distribution in the lower planetary boundary layer - drone-based investigation	*Atmos Chem Phys*	Field measurement	Drone vertical profiles of particle mass and number, CO2, O3 to 500 m	Germany (rural)
Dust lidar asymptotics	Asymptotic behavior of lidar scattering properties of absorbing dust aerosols across Rayleigh and geometrical-optics regimes: theory and implications	*Atmos Chem Phys*	Theoretical / optical modelling	Lidar ratio, depolarisation ratio, SSA vs size parameter and iron-oxide fraction	Not applicable (theory)
Taiwan transboundary PM_{10} forecast	Forecasting transboundary PM_{10} pollution in northern Taiwan using upwind observations and transport timing	*Environ Pollut*	Machine-learning model	PM_{10} at 3 receptors; 9 continental upwind records; trajectory timing composite	Taiwan (northern)
Exposure assessment & modelling					
Hanoi $PM_{2.5}$-bound PCDD/Fs	Levels and contributing sources to $PM_{2.5}$-bound PCDD/Fs and dl-PCBs in ambient air of Hanoi	*Atmos Pollut Res*	Metadata only (no abstract)	$PM_{2.5}$-bound PCDD/Fs and dioxin-like PCBs	Vietnam (Hanoi)
North Macedonia biomass PMF	Biomass burning drives urban $PM_{2.5}$ across the Western Balkans: evidence from a six-city source apportionment study in North Macedonia	*Environ Pollut*	Source apportionment	$PM_{2.5}$ from 1,195 daily filters, EPA PMF 5.0	North Macedonia (six cities)
Sao Paulo 2024 fire extremes	Compound fire-meteorology extremes drove megacity air pollution during the 2024 South American fire season	*J Environ Manage*	Chemical transport model	Surface $PM_{2.5}$ and CO, AOT, trajectories; fire-on/fire-off global simulations	Brazil (Sao Paulo State)
US school wildfire smoke trends	Spatio-Temporal Trends in Wildfire Smoke Exposures at Schools and Childcare Centers in the United States and Implications for Child and Adolescent Health	*Environ Res*	Metadata only (no abstract)	Wildfire smoke $PM_{2.5}$ at school and childcare locations	United States

First author	Focus	Journal	Design	Metric	Region
Burden, policy & mitigation					
Canada wildfire smoke guidance	Public health guidance about wildfire smoke: Four targets for improvement	*Can J Public Health*	Commentary	Wildfire smoke (guidance)	Canada
Other clinical endpoints					
China fire-sourced $PM_{2.5}$ cancer	Extreme, Accumulated, and Lagged Exposure to Fire-Sourced $PM_{2.5}$ and Site-specific Cancer Risk in China	*Environ Pollut*	Ecological / registry analysis	Fire-sourced vs all-source $PM_{2.5}$; lagged, accumulated and extreme metrics	China (nation-wide)

Provenance and method

Entry window **13 October 2026**, single day. Harvester legs: **PubMed** health **14**, sensing **1**; **Europe PMC 46**, all AGRICOLA retro-index records (pubYear before 2026), **all dropped**; **Crossref by ISSN 42** (2 ES&T Letters front-matter items dropped); **arXiv** no entry dated 13 October; **OpenAlex** HTTP 429. 53 unique, 0 already carried. **PubMed connector** 8 PMIDs dated 13 October, 3 appended — **56** candidates screened, **16 admitted**, **40 rejected** with logged reasons (one a cross-day duplicate of the 12 October Lancet Planetary Health record). Abstract retry recovered 11 of 15; **4 records carried on metadata alone**. **Scholar Gateway** still `ACCESS_DENIED`. Abstracts remain the copyright of their publishers.